

Light – Reflection and Refraction (Set B) — MCQ Worksheet

Science • Chapter 9 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. A plane mirror always forms image that is:

- (A) Virtual, erect, of same size
- (B) Real, inverted
- (C) Magnified
- (D) Diminished

Q2. Focal length of plane mirror is:

- (A) Infinity
- (B) Zero
- (C) Equal to its size
- (D) Negative

Q3. Magnification by a plane mirror is:

- (A) 1
- (B) 0
- (C) -1
- (D) Variable

Q4. A concave mirror with focal length 10 cm has radius of curvature:

- (A) 20 cm
- (B) 10 cm
- (C) 5 cm
- (D) 40 cm

Q5. Object at focus of concave mirror gives image at:

- (A) Infinity
- (B) Focus
- (C) Centre
- (D) Pole

Q6. Object beyond C of concave mirror gives image:

- (A) Between F and C, real, inverted, diminished
- (B) Beyond C, magnified
- (C) At F
- (D) Virtual

Q7. Convex mirror is used as rear-view mirror because it gives:

- (A) Wider field of view; erect diminished images
- (B) Magnified images
- (C) Real images
- (D) Inverted images

Q8. Refractive index of glass relative to air is approximately:

- (A) 1.5
- (B) 0.5
- (C) 1
- (D) 2.5

Q9. Refractive index of water is approximately:

- (A) 1.33
- (B) 1.5
- (C) 2
- (D) 1

Q10. When light enters a denser medium, its speed:

- (A) Decreases
- (B) Increases
- (C) Stays same
- (D) Becomes infinite

Q11. Frequency of light when it goes from one medium to another:

- (A) Remains same
- (B) Doubles
- (C) Halves
- (D) Becomes zero

Q12. A convex lens of focal length 20 cm has power:

- (A) +5 D
- (B) +2.5 D
- (C) +10 D
- (D) +50 D

Q13. Concave lens has power:

- (A) Negative
- (B) Positive
- (C) Zero
- (D) Infinity

Q14. Real image is one which:

- (A) Can be obtained on a screen
- (B) Always virtual
- (C) Always erect
- (D) Cannot be seen

Q15. Sign convention for distances measured opposite to incident light:

- (A) Negative (B) Positive
(C) Zero (D) Variable

Q16. A concave mirror of focal length 15 cm produces an image at 30 cm in front of mirror. The object is at:

- (A) 30 cm (using $1/v + 1/u = 1/f$ with $f = -15$, $v = -30 \rightarrow u = -30$) (B) 10 cm
(C) 45 cm (D) 20 cm

Q17. A convex lens forms image at 15 cm for object at 30 cm. Focal length is:

- (A) 10 cm (B) 5 cm
(C) 15 cm (D) 20 cm

Q18. An object 4 cm tall is placed at 10 cm in front of concave mirror of focal length 5 cm. Image is real, at 10 cm and height:

- (A) 4 cm ($m = -1$) (B) 8 cm
(C) 2 cm (D) 16 cm

Q19. If image is virtual, erect and magnified, the mirror is:

- (A) Concave (object between F and pole) (B) Convex
(C) Plane (D) None

Q20. Two convex lenses of power +5 D and -2 D are placed in contact. Net power is:

- (A) +3 D (B) +7 D
(C) -7 D (D) +10 D

Q21. If a convex lens forms image of same size as object, object is at:

- (A) 2F (B) F
(C) Infinity (D) Between O and F

Q22. A pencil dipped in water appears bent at the surface because of:

- (A) Refraction of light at water-air boundary (B) Reflection
(C) Absorption (D) Dispersion

Q23. A glass slab causes:

- (A) Lateral shift of light (no angular deviation) (B) Bending of light without shift
(C) Strong reflection (D) Total internal reflection

Q24. Speed of light in vacuum is:

- (A) 3×10^8 m/s (B) 3×10^6 m/s
(C) 3×10^{10} m/s (D) Variable

Q25. The magnification of a lens is +2. This implies:

- (A) Image is virtual, erect and twice the size (B) Image is real and inverted
(C) Image is diminished (D) Image is upside-down